Constellation Debris Detection Trade Study V2.5

Model Methodology and User Guide
Senior Design Working Document

Sean Elliott

November 1, 2026

Generative AI Disclosure: This working document was created with assistance from generative AI using the MATLAB script, and input files.

Purpose and version scope. This document describes the V2.5 constellation trade-study script, `constellationDetection.m`, using the final review snapshot supplied as `constellationDetection(1).m` on November 1, 2026 [1].

The model estimates how often debris makes a *qualified passage* through a sensor's observable field of view (FOV). A passage qualifies when its estimated visible time is at least the minimum residence time supplied by the sensor team. MASTER provides the debris density; a statistical orbit-orientation model provides encounter velocities; geometric integration combines these with the sensor response. Outputs are expected passages per day and over the mission, by size bin, for a mean spacecraft and the constellation.

A passage is not a camera pixel event, a data-volume estimate, or a uniquely identified object. The same object can contribute multiple passages. Fractional expected counts are valid statistical outputs.

1 Assumptions and Limitations

V2.5 is an architecture-level model. It does not propagate individual debris objects or reproduce the full optical/event-generation process. Its principal assumptions are below.

Assumption / limitation	Rationale and effect
Circular spacecraft and debris orbits	Speeds depend only on altitude. The spacecraft has no radial velocity. Eccentric motion, orbital decay, and individual debris trajectories are not modeled.
First-order SSO design	The averaged J_2 nodal rate sets Sun-synchronous orbit (SSO) inclination. Higher-order gravity and other perturbations are omitted. Representative positions replace time-resolved propagation.
Altitude-only MASTER environment	One differential altitude–diameter density table supplies human-made debris. Full declination and right-ascension coverage is required. Density is treated as spatially uniform on each altitude shell; no declination correction is applied.
Fixed population snapshot	The supplied environment is referenced to 2024/08, while the default mission starts in 2027. Population density is held fixed throughout the mission. Counts are scenario extrapolations, not forecasts of debris evolution.
Independent inclination sampling	Debris inclination is uniform in angle by default, relative RAAN is uniform, and both node-crossing signs have equal weight. Uniform inclination is not isotropic orientation. These weights and MASTER density are not a jointly fitted phase-space distribution.
Local velocity approximation	Debris velocities lie in the spacecraft's local tangential–normal plane at modeled plane crossings. This does not reconstruct the velocity distribution at every observed spatial location.
Straight observable chords	Each crossing uses constant local relative velocity and fixed local pointing. Orbital curvature, attitude slew, and changing velocity during a crossing are neglected.
Midpoint evaluation	Density, sensor threshold, and optional phase angle are evaluated at the midpoint of an observable chord. Their variation along that chord is not integrated.
Representative diameter	Each bin uses its full imported population density but only one configurable diameter for sensor response. This can bias counts when response changes strongly across a bin.
Sensor-provided illumination	The CSV is assumed to include the relevant illumination and sensor physics. No terminator cosine, albedo multiplier, or additional shadow penalty is applied. A fixed-phase response is not valid at all real mission geometries.
Earth and Sun masks	Earth-background directions are removed, including foreground debris against Earth. Sun proximity blocks the entire FOV; the square-FOV check is conservative. Earth and solar geometry are simplified.
Uniform operating availability	Communications downtime reduces all sampled rates by the same factor. Its possible correlation with illumination, pointing, or orbital location is not modeled.
Constellation scaling	Spacecraft counts scale passages linearly within each plane. No identity tracking, spacecraft phasing, or observation-correlation model is included. Unique/repeat estimates have been removed.

Assumption / limitation	Rationale and effect
Altitude capability	Profiles apply an exact sensor diameter to the density of its associated finite size bin. They are conditional passage fractions, not shell-volume coverage or exact-diameter population counts. A single positive-weight qualifying sample can set the reported maximum altitude.
Numerical sampling	Finite area, velocity-altitude, orientation, date, and orbital-position samples approximate continuous averages. Default settings are starting values, not demonstrated convergence.

2 User Inputs

Normal configuration is grouped in `USERINPUTS` at the top of the script. The following values describe the saved V2.5 file, not recommended final mission requirements. Paths are relative to the script folder unless changed.

Mission, orbit, and population

Input	Current setting and meaning
`Mission.StartDate`, `Mission.Duration_days`	January 1, 2027; $5 \times 365.25 = 1826.25$ days. Dates affect Sun geometry, not MASTER population evolution.
`Shell.MinAltitude_km`, `Shell.MaxAltitude_km`	600 and 1000 km geocentric altitude above the model's spherical Earth.
`Orbit.Altitude_km`, `Orbit.Type`	600 km; `SSO`. Alternative `Circular` uses `Orbit.Inclination_deg`.
`Orbit.NominalLTAN_hr`	18 hours, corresponding to the nominal dusk ascending node and dawn descending node.
`Orbit.NumPlanes`, `Orbit.SpacecraftPerPlane`	3 planes, 9 spacecraft per plane (27 spacecraft total).
`Orbit.BaselineLTANSpread_min`	15-minute total span; planes distributed symmetrically about nominal LTAN.
`Debris.DensityModel`	`MASTER`. `Demonstration` instead uses explicitly synthetic per-bin densities and labels results accordingly.
`Debris.MASTERFolder`	`MASTER_Data`; searched recursively.
`Debris.MASTERFileName`	`ODIN_2024_D010_070_t.__1`. Exactly one matching file is required. Blank enables header-based discovery of exactly one suitable table.
`Debris.ExpectedReferenceEpoch`	Blank by default; can require an exact metadata match such as `2024/08`.
`Debris.ScienceSizeRange_cm`	1–3.1 cm in this final configuration. Bins must not straddle a science boundary.
`Debris.MaxUsableDiameter_cm`	5 cm. Data beyond this limit are not counted even if present in the export.
`Debris.InclinationModel`	`Uniform`; `Function` accepts nonnegative weights with a positive total from `Debris.InclinationFunction`.

Diameter bin (cm)	Representative diameter (cm)	Classification
1–1.5	1	Science
1.5–2.5	2	Science
2.5–3.1	3	Science
3.1–5	5	Outside science

The arrays `Debris.SizeBins_cm` and `Debris.RepresentativeDiameter_cm` set these choices. Adjacent bins use disjoint MASTER cells, so shared boundaries are not double-counted. A representative diameter at a bin boundary does not change the density assigned to that bin.

Sensor, pointing, and operations

Input	Current setting and meaning
`Sensor.CSVFile`	`Sensor_Residence_Time_Trade2.csv`; change this path when using a different sensor response.
`Sensor.FOVShape`	`Square` in the saved V2.5 file. Set `Circular` for a circular cone.
`Sensor.UseCSVFOV`, `Sensor.FOV_deg`	CSV FOV enabled. `HFOV_deg` is interpreted as full width: each side for square, full cone angle for circular. Manual 1.4-degree value is inactive unless CSV FOV is disabled.
`Sensor.ExpectedAlbedo`	0.20; checks CSV metadata, not a brightness scaling factor.
`Sensor.UsePhaseAngleCSV`	False. True requires a complete multi-phase CSV grid.
`Sensor.FixedPhaseAngle_deg`	30 degrees when phase interpolation is disabled.
`Sensor.ReportPhaseDiagnostic`	False; optionally reports sampled chord-midpoint phase extrema.
`Pointing.Mode`, `Pointing.CustomRTN`	`CustomRTN`; $[\sin 66^\circ, 0, \cos 66^\circ]$. Other modes: Zenith, AlongTrack, Normal+, Normal-.
`Pointing.ChooseAntiSunNormal`	False; the positive normal component in CustomRTN remains fixed. If enabled, the code selects the normal-component sign opposite the Sun while preserving the radial and tangential components.
`Comms.Downlink_min_per_day`	120 minutes total per day across all sessions.
`Comms.Transition_min_per_session`, `Comms.Sessions_per_day`	10 minutes total overhead per session; one session per day. This is the time required to transition to comms mode and point the antenna. The overhead is not applied separately at each end. Placeholder value.
`Exclusion.SunClearance_deg`, `Exclusion.EarthLimbClearance_deg`	Extra angular margins of 5 degrees and 0.05 degrees, respectively. Reasonable limits, subject to change.

Altitude capability inputs

`Capability.Enabled=true` adds baseline altitude profiles for `Capability.Diameters_cm=[1235]`. `Capability.AltitudeStep_km=10` sets reporting intervals, independently of the projected-area grid and velocity-altitude intervals. These diagnostics do not change the main size-bin rates. In this snapshot, diagnostic diameters and rate-bin representatives happen to coincide.

The final science interval is **1–3.1 cm**, with the 2.5–3.1 cm population evaluated using a 3 cm response. This differs from the earlier 1–3 cm configuration: objects in the added 3–3.1 cm interval contribute to the reported science total. The documentation follows this explicit code setting without assuming a reason for the change.

Numerics, trades, and output controls

- Orientation resolution: `Debris.NumInclinationSamples`=100 and `Debris.NumRAANSamples`=8; two node signs are included.
- Velocity-altitude spacing: `Numerics.VelocityAltitudeStep_km`=10. This is distinct from the MASTER density-cell width.
- Area integration: `Numerics.ProjectedAlongSamples`=160 and `Numerics.ProjectedAcrossSamples`=24.
- Geometry sampling: `Numerics.OrbitalPositionSamples`=12 and `Numerics.MissionDateSamples`=24. Both use equally weighted midpoint samples.
- `Numerics.RunGeometrySelfChecks`=true enables small analytical geometry checks before the mission calculation.
- `Trade.RunArchitectureTrade`, `Trade.RunLTANTrade`, and `Trade.RunAltitudeTrade` are all false by default. The configured ranges are 1–6 planes, 1–9 spacecraft per plane, LTAN spreads of 0–180 minutes in 10-minute steps, and spacecraft altitudes of 570–700 km in 10 km steps. The architecture trade uses a 60-minute total LTAN spread.
- `Plot.MakeFigures`=true, `Plot.SaveFigures`=true, and `Output.SaveTables`=true. `Output.ReportPoissonProbability`=false by default.

3 Model Methodology and Governing Equations

3.1 Circular Orbit and SSO Geometry

The script uses $R_E = 6371$ km, $\mu = 398600$ km^3/s^2, and $J_2 = 1.08262668 \times 10^{-3}$. These are the constants in this script; they should not be confused with constants in the CDH pipeline interface. For altitude h_s, circular radius, speed, and mean motion are

$$r_s = R_E + h_s, \qquad v_s = \sqrt{\mu/r_s}, \qquad n_s = \sqrt{\mu/r_s^3}. \tag{1}$$

The first-order nodal rate is

$$\dot{\Omega} = -\frac{3}{2} J_2 n_s \left(\frac{R_E}{r_s}\right)^2 \cos i_s. \tag{2}$$

For SSO, the script solves for i_s using a positive nodal rate of one revolution per 365.2422 days and rejects an impossible cosine. Circular mode instead uses the selected inclination. These relations set the orbit design; they do not propagate a full perturbed trajectory.

The local RTN frame has $\hat{\boldsymbol{R}}$ radially outward, $\hat{\boldsymbol{T}}$ along circular motion, and $\hat{\boldsymbol{N}} = \hat{\boldsymbol{R}} \times \hat{\boldsymbol{T}}$. In this frame the spacecraft is at $[r_s, 0, 0]$ relative to Earth and has velocity $[0, v_s, 0]$.

3.2 Debris Shell and Population

This version imports *spatial density*, not impact flux. The selected MASTER output [3] contains a differential altitude–diameter grid in objects/km^3, with human-made debris selected. Its header identifies a 2024/08 reference population and a simulation interval at 2024/07/31.99. The current export covers 600–1000 km and 1–7 cm, over all declinations and right ascensions. Only the configured 1–5 cm bins are used.

The importer reads axis descriptions and units from the header, supports either axis order, and requires a complete uniform linear grid without duplicate, missing, negative, or nonfinite cells. It checks the human-made column against the total to reject meteoroid contributions. This parser expects the supplied MASTER format with two coordinate columns and eighteen density-source columns; it is not a universal importer for arbitrary MASTER exports.

Let ρ_{jk} be the reported density in altitude cell j and diameter cell k. For configured size bin b,

$$\rho_{jb} = \sum_{k \in b} \rho_{jk}. \tag{3}$$

The supported export supplies density per diameter cell in objects/km^3. Therefore cells are summed directly: **no second multiplication by diameter width is applied**. Diameter values are converted from meters to centimeters. Bin edges must match imported cell edges; partial-cell allocation and size extrapolation are rejected.

Density queries use a piecewise constant altitude cell value, $\rho_b(h) = \rho_{jb}$ for h in cell j. No declination or inclination multiplier is applied to density. Inclination weights affect velocity averaging separately.

For diagnostics only, the exact intersected shell-cell volume and volume-weighted mean density are

$$V_j = \frac{4\pi}{3}\left[(R_E + h_{j,o})^3 - (R_E + h_{j,i})^3\right], \qquad \bar{\rho}_b = \frac{\sum_j \rho_{jb} V_j}{\sum_j V_j}. \tag{4}$$

These means help inspect the imported population; the encounter calculation uses local altitude-cell density, not a shell-average replacement.

3.3 Debris Orientation and Relative Velocity

Inclinations i_d are sampled from 0 to 180 degrees, relative right ascension of the ascending node (RAAN) $\Delta\Omega$ from 0 to less than 360 degrees, and both node signs $\sigma = \pm 1$ are included. The mutual plane angle satisfies

$$\cos\Phi = \cos i_s \cos i_d + \sin i_s \sin i_d \cos\Delta\Omega. \tag{5}$$

At a velocity-altitude interval midpoint h_j^v,

$$\boldsymbol{v}_d = \sqrt{\frac{\mu}{R_E + h_j^v}}[0, \cos\Phi, \sigma\sin\Phi], \quad \boldsymbol{v}_{\rm rel} = \boldsymbol{v}_d - [0, v_s, 0], \quad v_{\rm rel} = \|\boldsymbol{v}_{\rm rel}\|. \tag{6}$$

For inclination weights w_i, RAAN count N_Ω, and either node sign, the normalized sample weight is

$$w_q = \frac{w_i}{\sum_i w_i}\frac{1}{N_\Omega}\frac{1}{2}, \qquad \sum_q w_q = 1. \tag{7}$$

Uniform mode uses equal w_i. Equivalent sampled orientations can be combined without changing their total weight. Zero-speed samples do not contribute swept volume.

These are local statistical velocities, not reconstructed trajectories through the sensor. In particular, circular debris at a different geocentric location need not have zero radial component in the spacecraft's RTN frame. V2.5 retains that simplifying plane-crossing approximation.

3.4 Sensor Detection Volume

Let $\boldsymbol{x}$ point from the spacecraft to a candidate debris location, $s = \|\boldsymbol{x}\|$ be slant range, and $\hat{\boldsymbol{b}}$ be the unit boresight. Full FOV width Θ gives half-angle $\beta = \Theta/2$. A circular FOV requires

$$\boldsymbol{x} \cdot \hat{\boldsymbol{b}} \geq s \cos \beta. \tag{8}$$

For a square FOV, choose perpendicular unit axes $\hat{\boldsymbol{a}}, \hat{\boldsymbol{c}}$ across the boresight. With $z = \boldsymbol{x} \cdot \hat{\boldsymbol{b}}$,

$$z \geq 0, \qquad |\boldsymbol{x} \cdot \hat{\boldsymbol{a}}| \leq z \tan \beta, \qquad |\boldsymbol{x} \cdot \hat{\boldsymbol{c}}| \leq z \tan \beta. \tag{9}$$

The square roll convention projects local T perpendicular to the boresight to define $\hat{\boldsymbol{a}}$, falling back to N when necessary; $\hat{\boldsymbol{c}} = \hat{\boldsymbol{b}} \times \hat{\boldsymbol{a}}$.

Both shapes are bounded by *true slant range*, $s_{\min} \leq s \leq s_{\max}$, taken from CSV coverage. If zero is tabulated, it is only a mathematical endpoint and is never a visible point. A positive minimum tabulated range becomes the actual lower bound of the modeled volume. There is no extrapolation beyond the farthest tabulated range. The range bound is a data-support limit, not a guarantee that a crossing qualifies there.

Earth occultation and background clearance

For spacecraft radius r_s, the apparent Earth angular radius is $\alpha_E = \arcsin(R_E/r_s)$. If $\hat{\boldsymbol{e}} = [-1, 0, 0]$ points toward Earth, a line of sight is retained only when

$$\arccos\left(\frac{\boldsymbol{x}}{s} \cdot \hat{\boldsymbol{e}}\right) > \alpha_E + \delta_E, \tag{10}$$

where δ_E is the limb margin. The geometric boundary clips observable chords. This removes occulted directions and also removes foreground objects against the Earth disk. It is consequently more conservative than checking only whether the segment to the debris intersects Earth.

3.5 Effective Encounter Area and Observable Residence Time

For each sampled velocity, the model places deterministic area elements ΔA_ℓ on a plane perpendicular to $\hat{\boldsymbol{v}}_{\text{rel}}$. Each represents a bundle of parallel straight crossing paths,

$$\boldsymbol{x}_\ell(t) = \boldsymbol{p}_\ell + t\hat{\boldsymbol{v}}_{\text{rel}}. \tag{11}$$

Here t is a distance coordinate, not elapsed time. The code intersects each line with the FOV boundaries, spherical range bounds, physical inner and outer debris-shell surfaces, and Earth exclusion boundary. It tests the intervals between intersections and merges adjoining valid intervals into complete observable chords.

For chord length L_ℓ, residence time is

$$\tau_\ell = L_\ell / v_{\text{rel}}. \tag{12}$$

Artificial altitude-layer boundaries do not split a passage or shorten its residence time. The chord midpoint selects the velocity-altitude interval and MASTER density cell. A real excluded gap can produce two disconnected observable intervals, which the model counts as two passages.

Area sampling uses a cubic coordinate spacing along the projected boresight direction to resolve short ranges near the sensor apex, with transverse strip samples and their actual area weights. This improves near-range resolution but does not eliminate the need for convergence testing.

Sensor CSV interpolation and qualification

The required columns are `DebrisSize_cm`, `SlantRange_km`, `MinResidenceTime_s`, and `PhaseAngle_deg`. The grid must contain every range–diameter–phase combination exactly once. FOV metadata are required when `UseCSVFOV` is true. Albedo metadata, when present, must match the configured expected value; missing albedo is flagged.

Linear gridded interpolation supplies $\tau_{\min}(s, d)$ at a fixed phase, or $\tau_{\min}(s, d, \alpha)$ with phase interpolation enabled. The current table provides 1, 2, 3, and 5 cm response curves; intermediate representatives use interpolation. For bin b and chord midpoint range s_ℓ,

$$Q_{\ell b} = \begin{cases} 1, & \tau_\ell \geq \tau_{\min}(s_\ell, d_b, \alpha_\ell), \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

Use the fixed phase instead of α_ℓ when the phase toggle is off. Out-of-support response queries with positive density stop the calculation; they are not silently extrapolated. The table's `MaxRelativeVel_kms` and `MaxAngularRate_degs` are not additional thresholds. Encounter velocities come from Section 3.3, and residence qualification already applies the selected sensor condition.

3.6 Encounter Rate

For one spacecraft geometry, bin b has qualified passage rate

$$\dot{N}_b \approx \sum_{j,q} \sum_{\ell \in \mathcal{C}_{jq}} \rho_b(h_\ell)\, v_{jq}\, \Delta A_{jq\ell}\, w_q\, Q_{jq\ell b}. \tag{14}$$

$\mathcal{C}_{jq}$ contains observable chords whose midpoint belongs to velocity interval j for orientation q. Density uses actual midpoint altitude h_ℓ; speed uses that interval's center. The units are

$$(\text{objects/km}^3)(\text{km/s})(\text{km}^2) = \text{objects/s}.$$

This is a swept-volume rate: projected area times relative speed gives the volume encountered per second. It is not instantaneous occupancy divided by an arbitrary sampling time. Counts reflect both population abundance and qualification geometry. A larger object can qualify at a longer range but still produce fewer passages if its size bin contains less debris.

Performance improvements in the final implementation

The physical chord construction and orientation weights are retained. Interval testing reuses path projections onto the boresight and square-FOV axes, calculates Cartesian components directly, and skips intervals already rejected by the boundary tests. These changes avoid repeated coordinate-matrix construction; they do not deliberately lower numerical resolution or introduce a different crossing model.

After geometry is built, `prepareResponseGeometry` stores chord ranges and density-weighted rate contributions. When phase interpolation is disabled, it evaluates the representative-diameter residence tests once per pointing geometry and stores the resulting rates. Orbital-position, date, plane, and architecture/LTAN evaluations reuse those values while still applying Sun exclusion and selecting the applicable pointing geometry. Fixed-phase capability numerators and denominators are also cached. Altitude trades rebuild geometry and these cached quantities at each spacecraft altitude.

With phase interpolation enabled, density and range information remain reusable, but the sensor threshold is reevaluated at each sampled phase. These optimizations are intended to preserve the numerical calculation, apart from possible floating-point roundoff near boundaries. No measured speedup or convergence certification is implied by their inclusion.

3.7 LTAN, Sun Geometry, Illumination, and Availability

For $N_p > 1$ planes, offsets span the selected total LTAN spread uniformly; a single plane receives zero offset. Each plane is sampled at midpoint dates across the mission and midpoint arguments of latitude around a circular orbit. No individual spacecraft phase history is generated.

The code uses a low-order geometric Sun direction based on days from J2000, a uniformly advancing mean solar longitude, and approximate obliquity. For SSO the sampled node is set by

$$\Omega(t) = \bar{\lambda}_{\odot}(t) + 15^{\circ}[\mathrm{LTAN}_{\mathrm{hr}} - 12]. \tag{15}$$

The implementation uses mean solar longitude as its mean-Sun RA convention. Circular non-SSO mode initializes this relation at mission start and advances RAAN with the first-order J_2 rate. These are approximate architecture geometries, not precision solar ephemerides or node-time solutions.

Sun exclusion and phase angle

Let $\hat{\boldsymbol{s}}$ be the unit direction toward the Sun in RTN. The whole sample is blocked when

$$\arccos(\hat{\boldsymbol{b}} \cdot \hat{\boldsymbol{s}}) \le \beta_{\mathrm{bound}} + \delta_{\odot} + 0.272^{\circ}. \tag{16}$$

The final term is the fixed conservative solar angular radius used in this model. For a circular FOV, $\beta_{\mathrm{bound}} = \beta$; for square, $\beta_{\mathrm{bound}} = \arctan(\sqrt{2}\tan\beta)$. The square is enclosed by a cone for this test, so some geometries outside the actual square are also excluded.

When requested, phase angle at a chord midpoint is

$$\alpha_{\ell} = \arccos\left[-\frac{\boldsymbol{x}_{\ell}}{s_{\ell}} \cdot \hat{\boldsymbol{s}}\right]. \tag{17}$$

This is the Sun–debris–sensor angle, using a distant-Sun approximation. It is not the RTN tilt. With phase interpolation off, the fixed CSV slice is used everywhere. Turning on phase diagnostics alone does not modify detection rates; its extrema describe all sampled observable chord midpoints after Sun exclusion, not just qualified passages.

There is no separate diffuse-sphere brightness calculation, Earth-shadow screening of debris, or albedo rescaling in this script. Illumination validity therefore depends on what the supplied response table represents. A phase sweep improves response evaluation but does not, by itself, determine whether debris is in Earth's shadow.

Communications availability

With daily downlink time D, transition overhead H per session, and n_c sessions per day,

$$f_{\text{obs}} = 1 - \frac{D + n_c H}{1440}. \tag{18}$$

The code requires total downtime between zero and one day. This factor is applied once after averaging the geometric rates. It assumes downtime is independent of viewing opportunity; no extra communications penalty is embedded in the CSV.

3.8 Mission Detection Statistics

Let $\overline{\dot{N}}_{pb}$ be the representative date/orbit average for plane p, with Sun-blocked samples contributing zero, before downtime. With N_s spacecraft per plane and T_d mission days,

$$N^{\text{day}}_{\text{mean SC,b}} = 86400\, f_{\text{obs}} \frac{1}{N_p} \sum_p \overline{\dot{N}}_{pb}, \tag{19}$$

$$N^{\text{day}}_{\text{const,b}} = 86400\, f_{\text{obs}} N_s \sum_p \overline{\dot{N}}_{pb}, \tag{20}$$

$$N^{\text{mission}}_b = T_d\, N^{\text{day}}_b. \tag{21}$$

The per-spacecraft result is an average across planes; individual plane values remain available. Science, outside-science, and all-modeled totals sum the corresponding disjoint bins. "All modeled" means only configured sizes, not every object that could produce sensor activity.

Unique-object and repeated-observation estimates are removed. The optional output

$$P(\text{at least one constellation passage in a day}) = 1 - e^{-\Lambda_{\text{day}}} \tag{22}$$

assumes a stationary Poisson passage process with mean Λ_{day}. It is not a per-object detection probability or a claim of independent real observations. Expected-count outputs do not require this Poisson option.

3.9 Altitude Capability Profiles and Sampled Altitude Limits

The altitude diagnostic answers: *At which debris altitudes do modeled observable passages satisfy the response for a selected diameter?* It does not estimate the fraction of the entire spherical shell seen by the constellation.

For each requested diameter d, the code selects a population bin $b(d)$ containing it. At a shared boundary it selects the first matching bin, which is the lower-diameter bin when bins are sorted. The current mappings are 1 cm to 1–1.5 cm, 2 cm to 1.5–2.5 cm, 3 cm to 2.5–3.1 cm, and 5 cm to 3.1–5 cm. Only the sensor response uses the exact diagnostic diameter; population density still represents the entire associated bin.

For an observable chord midpoint $\boldsymbol{x}_\ell$ in the spacecraft frame, debris altitude is

$$h_\ell = \|[r_s, 0, 0] + \boldsymbol{x}_\ell\| - R_E. \tag{23}$$

This is geocentric altitude above the spherical Earth, not slant range or height above the spacecraft. The full observable chord sets residence time; the midpoint sets the altitude interval and response-query range.

Let $\mathcal{A}_k$ contain sampled observable chords with midpoint altitudes in reporting interval k, summed over the baseline plane/date/orbital-position samples that pass Sun exclusion. Let

$$w_{\ell d} = \rho_{b(d)}(h_\ell)\, v_\ell\, \Delta A_\ell\, w_{q(\ell)}. \tag{24}$$

The stored geometric and qualified weights and plotted fraction are

$$D_{kd} = \sum_{\ell \in \mathcal{A}_k} w_{\ell d}, \qquad U_{kd} = \sum_{\ell \in \mathcal{A}_k} w_{\ell d} Q_{\ell d}, \tag{25}$$

$$f_{kd} = \begin{cases} U_{kd}/D_{kd}, & D_{kd} > 0, \\ \text{NaN}, & D_{kd} = 0. \end{cases} \tag{26}$$

Here $Q_{\ell d}$ uses the exact requested diameter in the CSV residence-time test. A fraction of 0.7 means 70 percent of the modeled observable passage weight in that interval qualifies under that response. It is not a 70 percent probability of detecting any arbitrary object in the full altitude shell. No direct comparison of absolute event abundance follows from fractions alone.

Zeros and gaps. A zero fraction means geometric passage weight exists but no passage qualifies. A NaN means there is no sampled geometric passage weight, so the fraction is undefined; MATLAB leaves a gap in the plotted line. No qualifying samples give a NaN maximum altitude in the summary. Sparse projected-area sampling, zero population density, or excluded geometry can cause empty bins. In particular, the cubic sampling places fewer points at long range, and assigning full chords by midpoint can leave narrow upper-altitude reporting intervals empty. A gap is not proof of a physical blind altitude layer.

Maximum qualified midpoint altitude. The summary reports

$$h^{\text{sampled}}_{\max,d} = \max_{\ell:\, w_{\ell d} > 0,\, Q_{\ell d} = 1} h_\ell. \tag{27}$$

This is the highest positively weighted qualifying chord midpoint among sampled viewing conditions. It is not the highest point along a chord, a maximum slant range, or a guarantee that all lower altitudes are detectable. The reported interval bounds merely bracket that sampled maximum. A low-weight favorable crossing can set the maximum; it must be considered alongside qualification fractions and absolute weights.

Summary field	Meaning
`Diameter_cm`	Exact sensor-response diameter.
`SpacecraftAltitude_km`	Baseline spacecraft altitude, currently 600 km.
`DensityBinIndex`	Index into `Debris.SizeBins_cm` used for population weighting.
`AnyQualifiedPassage`	True if at least one positive-weight sampled passage qualifies.
`MaxQualifiedMidpointAltitude_km`	Highest qualifying midpoint altitude; NaN if none qualifies.
`HighestIntervalLower_km`, `HighestIntervalUpper_km`	Reporting-bin edges containing that maximum; NaN if none qualifies.

Inspecting absolute passage weight

The unnormalized arrays are `Diagnostics.CapabilityDenominator` (D) and `Diagnostics.CapabilityNumerator` (U). Rows index altitude intervals; columns follow `Capability.Diameters_cm`.

Divide either array by $N_p N_{\text{dates}} N_{\text{orbit}}$ to obtain a mean per-spacecraft equivalent rate in passages/s, then multiply by 86400 for passages/day. Multiplying additionally by f_{obs} includes scalar communications downtime. These are equivalent finite-bin population rates evaluated at the diagnostic diameter, not counts of objects of exactly that diameter. In this snapshot, representatives match diagnostic diameters, so summing the qualified altitude rates for a diameter should reproduce its corresponding mean-spacecraft size-bin rate, with consistent downtime treatment and roundoff tolerance.

Common normalization, spacecraft-count scaling, and uniform availability cancel in the ratio U/D. The capability fraction is therefore formed before communications downtime. Absolute weights remain in `Diagnostics` in the MAT output; the capability CSV contains fractions and flags, not these absolute rate arrays. Profiles are for the baseline architecture; optional trade tables do not export a separate profile for every trade case.

4 Using the Model

1. Place the script and selected sensor CSV together, or set the CSV path explicitly. Put the broad MASTER export in `MASTER_Data` or a configured subfolder. Individual file selection is not required at each run.
2. Select the intended MASTER filename. Blank discovery is useful only when exactly one compatible altitude–diameter table covers the requested shell and sizes. Multiple copies of the configured filename or multiple suitable candidates cause an error rather than an arbitrary choice.
3. Check header epoch, selected population, units, angular coverage, and numerical grid. Review `PopulationSummary` for zeros or unexpected size behavior before interpreting detection rates. The loaded-grid success message is not a population-validation statement.
4. Set bins and representative diameters. Align bin boundaries to MASTER cell edges and the science boundaries. To extend beyond 5 cm, deliberately revise the usable limit and bins, then verify environment and sensor coverage.
5. Select FOV shape, CSV FOV convention, pointing, margins, communications assumptions, orbit, and mission duration. Confirm that the CSV corresponds to the intended sensor/illumination conditions. A manual FOV change can invalidate its response even if the geometry still runs.
6. Keep phase interpolation off when using a single-phase CSV; verify that the selected file supports the configured 30-degree response slice. Enable it only with a complete grid covering all queried phases; optionally inspect diagnostic phases first.
7. Start with trade switches off. Run the baseline, inspect the density summary, input warnings, size-bin counts, and plane results. Demonstration-mode results must remain labeled synthetic.
8. Review `AltitudeCapability` and `AltitudeCapabilitySummary`, including NaN intervals, maximum-altitude interpretation, and absolute diagnostic weights. Repeat with finer area, orientation, velocity-altitude, date, and orbital-position sampling. Compare each size-bin rate, not just the total. Select and document an acceptable convergence tolerance before claiming quantitative performance.
9. Enable the desired trades after the baseline is reviewed. Retain the input CSV and MASTER export alongside the timestamped outputs for reproducibility.

The final area settings are 160 along-projection samples and 24 across-projection samples. Increased resolution alone does not demonstrate convergence. For capability-profile checks, keep the 10 km

reporting width fixed while refining area sampling; narrowing the reporting width without adding samples can increase NaN gaps. Compare bin-wise absolute rates and fractions as well as totals and sampled maxima. Do not replace gaps with zeros or interpolate through them solely to make a smoother figure.

The built-in geometry checks compare projected areas for simple axial and transverse circular-cone crossings and verify an Earth-facing exclusion case. They are regression checks on selected geometry conventions, not evidence of full-model convergence or environmental/optical validity.

Saved results

With `Output.SaveTables=true`, each run receives a timestamped subfolder under `model_results`.

File	Contents
`size_bin_passages.csv`	Bin limits, representative diameter, science flag, daily and mission passage counts per mean spacecraft and constellation; environment label.
`summary_passages.csv`	Science, outside-science, and all-modeled totals; optional daily Poisson probability.
`plane_passages.csv`	LTAN offsets and per-spacecraft daily rates for every bin and plane.
`imported_population_summary.csv`	Density minima, medians, maxima, volume-weighted means, zero-altitude-cell counts, and MASTER diameter cells summed per bin.
`architecture_trade.csv`, `LTAN_trade.csv`, `altitude_trade.csv`	Written only for enabled trades; daily science, outside-science, and total constellation passages.
`altitude_capability.csv`	Diameter, altitude-bin bounds and midpoint, qualification fraction, and observable/qualified-passage flags; written when capability diagnostics are enabled.
`altitude_capability_summary.csv`	Baseline spacecraft altitude, density-bin mapping, qualification flag, highest qualifying midpoint altitude, and its reporting interval for each diagnostic diameter.
`run_inputs_and_results.mat`	Saved configuration structures, baseline results, environment, population summary, diagnostics, capability settings and tables, and run label. Preserve the original CSV separately; the response grid is not included in this saved-variable list.
`RUN_STATUS.txt`	Environment label, sensor path, MASTER source path when applicable, phase toggle, and count/snapshot qualifications.

5 Generated Figures

The baseline figure is produced when `Plot.MakeFigures=true`. Trade figures also require their respective trade switches. There are no unique-object or repeat-observation figures in V2.5.

Figure	Content
Qualified object passages	Bars by size bin comparing mean-spacecraft and whole-constellation passages/day.
Altitude capability	Exact-diameter qualification fraction versus debris chord-midpoint altitude for 1, 2, 3, and 5 cm. NaN bins appear as gaps; connecting lines are visual guides, not continuous detection boundaries.
Architecture trade	Heat map of total constellation passages/day versus planes and spacecraft per plane. Plane/LTAN geometry is reevaluated; spacecraft count scales linearly.
LTAN trade	Science and outside-science constellation passages/day versus total LTAN spread. Effects arise from sampled solar geometry, sign selection, exclusion, and optional phase response, not a cosine multiplier.
Altitude trade	Science and outside-science constellation passages/day versus spacecraft altitude. Orbit and crossing geometry are rebuilt at each altitude; SSO inclination is recalculated when selected.

In the current implementation, `Plot.SaveFigures` exports the baseline bar chart as `qualified_passages.png` and the capability figure as `altitude_capability.png`. Figure saving is enabled in the final configuration, at 300 dpi in `trade_study_figures`. The trade and capability plots explicitly set white backgrounds and black axis/title/legend text; the baseline bar-chart text retains its existing MATLAB styling. The trade-plot helper displays its figures but does not automatically export them. Saved trade tables remain available for separate plotting.

References

[1] S. Elliott, *Constellation Debris Detection Trade Study*, MATLAB script `constellationDetection.m` (supplied as `constellationDetection(1).m`), V2.5-derived final design-review snapshot supplied November 1, 2026. Primary implementation source for this document.

[2] S. Elliott, *Constellation Debris Detection Trade Study V2.2: Model Methodology and User Guide*, Senior Design Working Document, October 2026. Supplied PDF; organizational basis for this revision.

[3] ESA MASTER v8.0.5 output, `ODIN_2024_D010_070_t.__1`, supplied altitude–diameter differential human-made spatial-density export; reference epoch 2024/08, datafile version 2408. Header and numerical format inspected for this revision.

[4] Hunter Liu, `Sensor_Residence_Time_Trade2.csv`, sensor response file selected by the final MATLAB snapshot. The configuration requests diameters 1, 2, 3, and 5 cm, albedo 0.20, and a fixed phase of 30 degrees. This selected CSV was not supplied with the final snapshot; its coverage, FOV, and optical metadata must be confirmed from the actual run inputs. Earlier response files are not assumed to have identical values.

[5] European Space Agency, *Software User Manual–MASTER*, revision 1.3, March 23, 2022, supplied project copy. Background documentation for MASTER configuration; the actual export header and format control the V2.5 importer.